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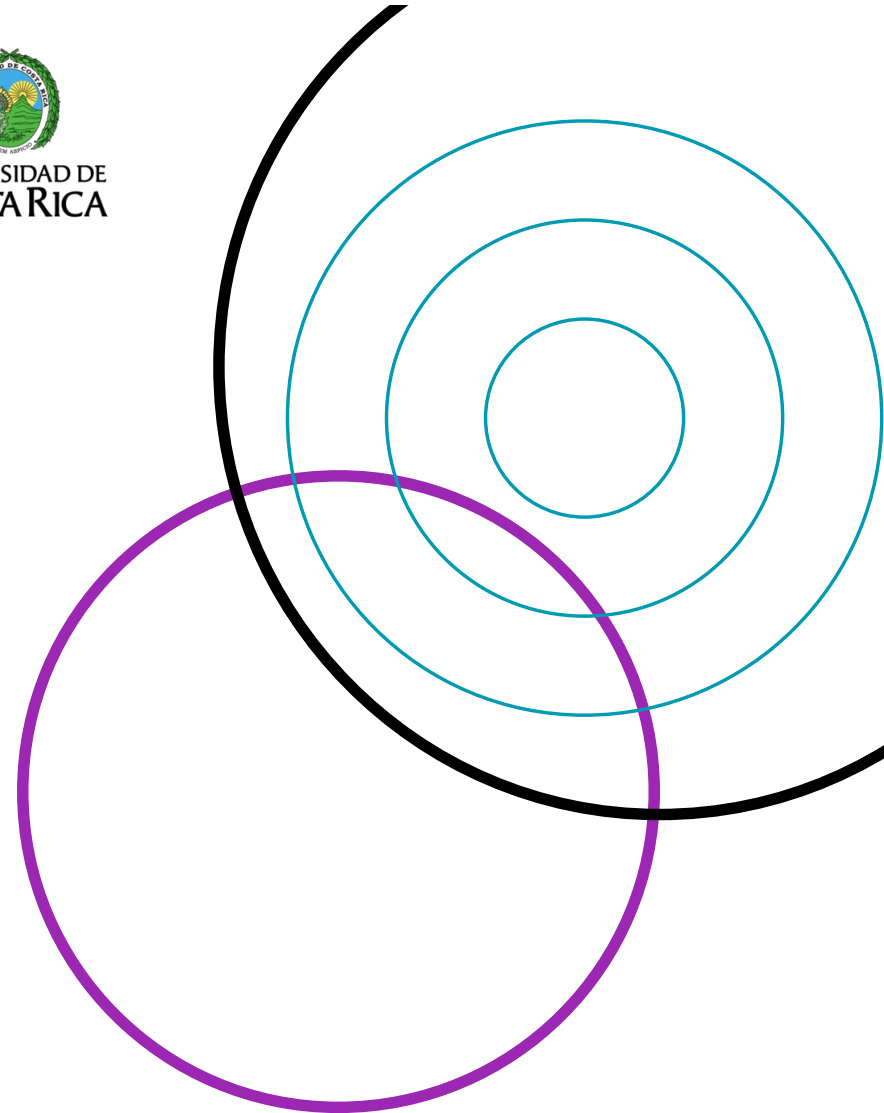


**UNIVERSIDAD DE
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Read Mapping

César Rodríguez

Bacterial Genomic Surveillance · 22–26 June 2026



OVERVIEW

Read mapping = entry door to genomic analyses

Sequencing gives us millions of DNA fragments. On their own they tell us nothing. Read mapping places each fragment back onto a known reference genome.

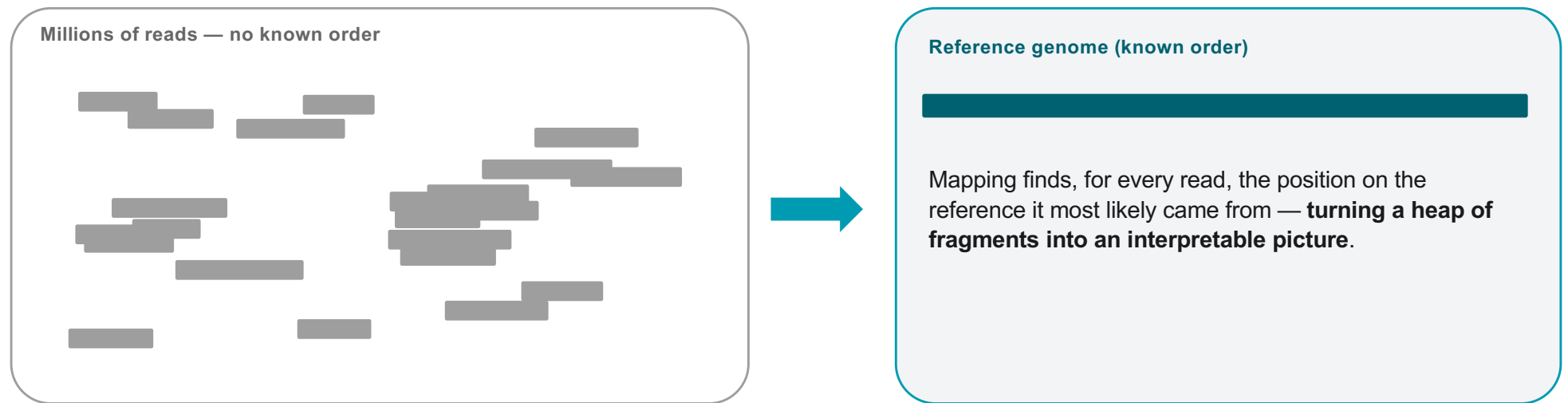


What this module covers

- Reads and the FASTQ file
- What mapping is, and why it is hard
- How mappers work
- The FASTQ → BAM → VCF workflow
- Strengths and limitations
- Real public-health applications

Sequencing produces thousands of anonymous reads

A sequencer fragments the DNA and reads pieces from random positions. Each read carries no information about where in the genome it came from — recovering that position is the whole job of mapping.



What is a read? The FASTQ file

Whatever the platform, every read arrives in the same 4-line text format. The fourth line — the quality string — tells you how much to trust each base.

1 Identifier	@M01234:55:000-ABCDE:1:1101:15589:1332 1:N:0:1
2 Sequence	GATCAGCAACGTACCGCCAGATACCGGGAACATACCATACGA
3 Separator	+
4 Quality	FFFFFFFFFFFFFF:FFFFFF,FFFFFF:FFFF,FF,F#FF##

Phred quality score $Q = -10 \cdot \log_{10}(P)$

Q30 = 1 error in 1,000 bases · Q20 = 1 in 100 · Q10 = 1 in 10. — FastQC plots this first.

Two classes of reads coexist

The kind of analysis you can do depends on the kind of reads you have.

Short reads — 2nd generation

- 100–300 bp (Illumina, the dominant platform)
- Very high accuracy (Q30+, <1 error / 1,000 bp)
- Massive throughput, low cost per base
- Paired ends help locate reads
- Too short to span long repeats

Long reads — 3rd generation

- 10–30 kb+ (PacBio HiFi, Oxford Nanopore)
- Span most repeats
- Portable/field-deployable (MinION on a laptop)
- Modern chemistries now reach Q20+
- Read methylation directly; also RNA and even proteins
- Higher cost per base (falling fast)

Complementary tools — many labs combine both.

WHAT MAPPING DOES

The core idea: place each read on a reference

Reads (FASTQ)



Reference genome (FASTA)



FASTQ



SAM / BAM

The mapper records each read's position, alignment quality and any mismatches.

THE CHALLENGE

Why it is hard: scale demands indexing

Comparing every read against every position of the genome the naïve way would take thousands of CPU-hours per sample.

100s of millions

reads per run

~4.6 Mbp

an *E. coli* genome

>3,000 Mbp

a human genome

1000s

CPU-hours if done naïvely

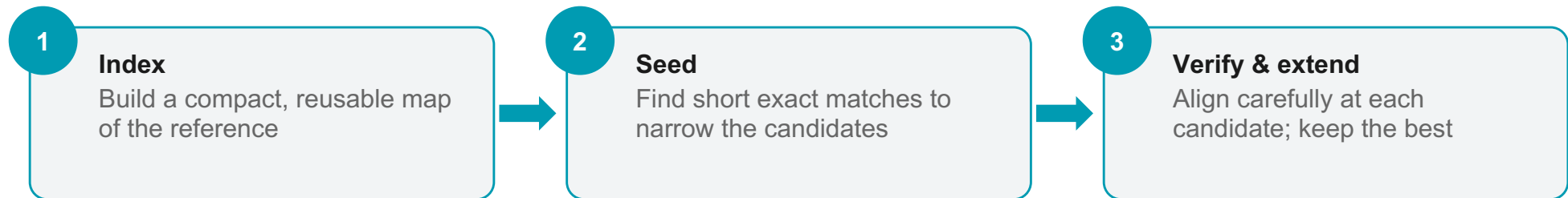
The breakthrough: indexing

Build a **compact, reusable data structure** of the reference once, then look up where any read belongs in essentially constant time.

THE CHALLENGE

How every mapper works: three steps

Mappers differ in detail, but all share the same strategy:



Indexing is done once and reused for every read

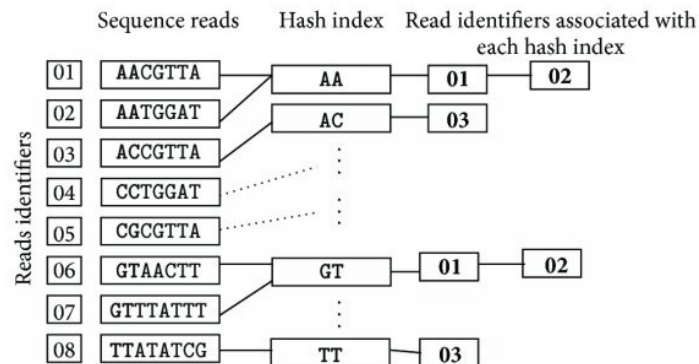
Verification uses classic base-by-base alignment (Smith–Waterman / Needleman–Wunsch) only at the few promising positions — never across the whole genome.

THE CHALLENGE

Two algorithm families

Hashing

- Store reference k-mers in a hash table of positions
- Look up the read's k-mers to get candidates
- Spaced seeds tolerate mismatches: with 4 seeds, one error still leaves ≥ 3 exact matches
- Maq, SOAP, BFAST



Burrows–Wheeler / FM-index

- Reversibly compress the reference, then search it backwards, character by character
- Human index ~ 2 GB — fits in a desktop's RAM
- $\sim 30\times$ **faster** than early hash mappers
- BWA, Bowtie2 — dominant since 2009

i	$T^{(i)}$		i	$T^{(i)}$		A	$T[A[j]..n]$
1	mississippi\$		12	\$mississippi		12	\$
2	ississippi\$m		11	i\$mississipp		11	i\$
3	ssissippi\$mi		8	ippi\$mississ		8	ippi\$
4	sissippi\$mis		5	issippi\$miss		5	issippi\$
5	issippi\$miss		2	ississippi\$m		2	ississippi\$
6	ssippi\$missi	sort	1	mississippi\$	$\hat{=}$	1	mississippi\$
7	sippi\$missis	$\Rightarrow$	10	pi\$mississip		10	pi\$
8	ippi\$mississ		9	ppi\$mississi		9	ppi\$
9	ppi\$mississi		7	sippi\$missis		7	sippi\$
10	pi\$mississip		4	sissippi\$mis		4	sissippi\$
11	i\$mississipp		6	ssippi\$missi		6	ssippi\$
12	\$mississippi		3	ssissippi\$mi		3	ssissippi\$

THE CHALLENGE

Long reads need a different strategy

Long reads are **noisier** and full of small insertions and deletions, so extending one anchor across the whole read (“seed-and-extend”) is slow and fragile. Modern long-read mappers instead use **seed-and-chain**.

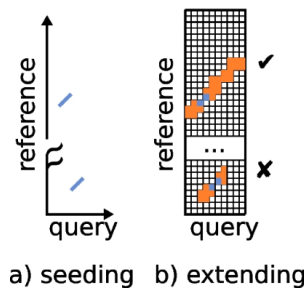
Seed-and-extend (short reads)

- One anchor, then extend by dynamic programming
- Quadratic time; intolerant of frequent indels
- Great for accurate short reads

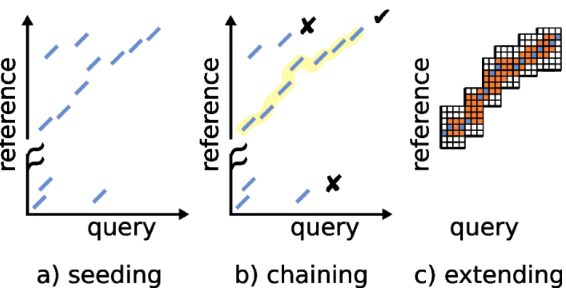
Seed-and-chain (long reads)

- Many anchors placed along the read
- Build and score a **chain of compatible anchors**
- Align only the gaps between them — minimap2

short-read mapping



long-read mapping



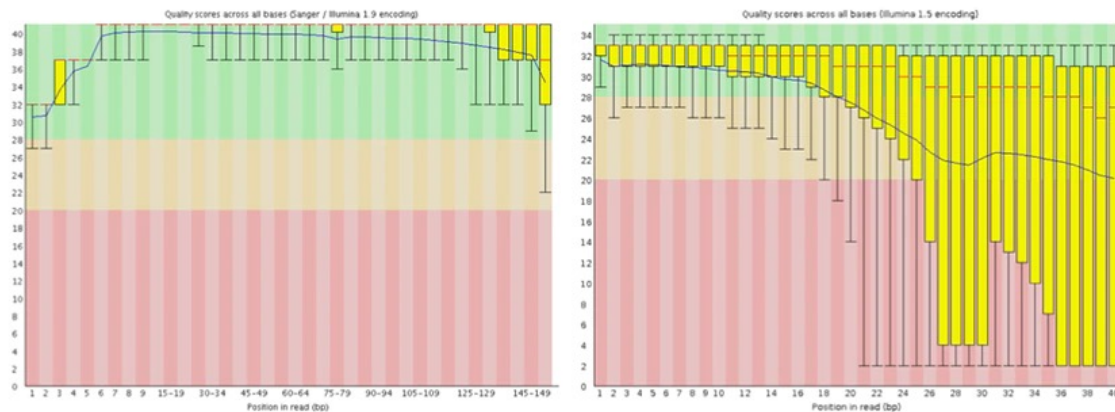
A standard reference-based workflow



Each stage hands a standard file to the next. The BAM file — sorted, compressed and indexed alignments — is the hub: it feeds variant callers, coverage tools, viewers, and structural-variant detectors.

Step 1 — Quality control and trimming

- **FastQC** inspects per-base quality profiles before anything else
- **Trimmomatic / fastp** remove sequencing adapters and low-quality ends
- **Quality shape** is high in the middle and falls towards the 3' end (illumina)
- **Species check** (i.e. Kraken2) catches contamination early



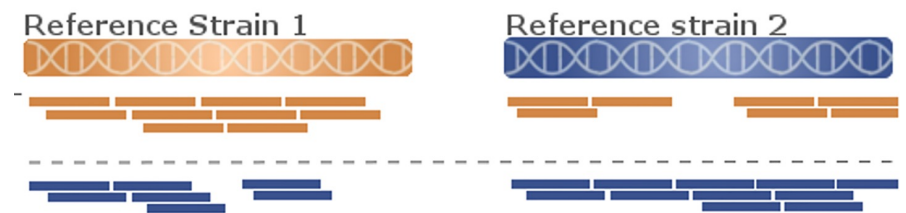
**Garbage in,
garbage out**

**Variant calls can only be as good as the
reads you feed the mapper.**

Step 2 — Choosing a reference

The **mapper only places reads on regions present in the reference**. Anything too divergent is silently discarded — so the more distant the reference, the more of the genome disappears from the analysis.

- **Best practice** — use the closest complete genome available
- **Ideally** the same species, and where possible the same clone or serotype
- **A closed, high-quality assembly** makes the cleanest coordinate system



Reference bias

A poorly chosen reference biases every downstream result — we return to this under limitations.

Step 3 — SAM/BAM: what an alignment stores

For every read, the SAM record answers: **where did it go**, and **how well does it fit?**

QNAME	read identifier	FLAG	pairing / orientation bits
RNAME + POS	reference contig and position	MAPQ	confidence the read is in the right place
CIGAR	matches, mismatches, insertions, deletions	SEQ + QUAL	the bases and their quality scores

SAM (text) → BAM (compressed, sorted, indexed)

BAM is the analysis-ready input for variant callers and the genome browsers (such as Artemis or IGV).

SAM format

Panel A: pileup view.

A

Coord	12345678901234	5678901234567890123456789012345
ref	AGCATGTTAGATAA**GATAGCTGTGCTAGTAGGCAGTCAGCGCCAT	
+r001/1	TTAGATAAAGGATA*CTG	
+r002	aaaAGATAA*GGATA	
+r003	gcctaAGCTAA	
+r004	ATAGCT.....TCAGC	
-r003	ttagctTAGGC	
-r001/2	CAGCGGCAT	

B

Header section

QHD VN:1.5 SO:coordinate
@SQ SN:ref LN:45

QUAL (read quality; * meaning such information is not available)

r001	99	ref	7	30	8M2I4M1D3M	=	37	39	TTAGATAAAGGATACTG	*
r002	0	ref	9	30	3S6M1P1I4M	*	0	0	AAAAGATAAGGATA	*
r003	0	ref	9	30	5S6M	*	0	0	GCCTAAGCTAA	* SA:Z:ref,29,-,6H5M,17,0;
r004	0	ref	16	30	6M14N5M	*	0	0	ATAGCTTCAGC	*
r003	2064	ref	29	17	6H5M	*	0	0	TAGGC	* SA:Z:ref,9,+,5S6M,30,1;
r001	147	ref	37	30	9M	=	7	-39	CAGCGGCAT	* NM:i:1

QNAME (query template name, aka. read ID)
FLAG (indicates alignment information about the read, e.g. paired, aligned, etc.)
RNAME (reference sequence name, e.g. chromosome /transcript id)
POS (1-based position)
MAPQ (mapping quality)
CIGAR (summary of alignment, e.g. insertion, deletion)
RNEXT (reference sequence name of the primary alignment of the NEXT read; for paired-end sequencing, NEXT read is the paired read; corresponding to the RNAME column)
PNEXT (Position of the primary alignment of the NEXT read in the template; corresponding to the POS column)
TLEN (the number of bases covered by the reads from the same fragment. In this particular case, it's 45 - 7 + 1 = 39 as highlighted in Panel A). Sign: plus for leftmost read, and minus for rightmost read
SEQ (read sequence)
Optional fields in the format of TAG:TYPE:VALUE

Panel B: raw SAM records with all 11 mandatory fields annotated

CIGAR string

B

Header section

QUAL (read quality; * meaning such information is not available)

```

@HD VN:1.5 SO:coordinate
@SQ SN:ref LN:45
r001 99 ref 7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG *
r002 0 ref 9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA *
r003 0 ref 9 30 5S6M * 0 0 GCCTAAGCTAA * SA:Z:ref,29,-,6H5M,17,0;
r004 0 ref 16 30 6M14N5M * 0 0 ATAGCTTCAGC *
r003 2064 ref 29 17 6H5M * 0 0 TAGGC * SA:Z:ref,9,+,5S6M,30,1;
r001 147 ref 37 30 9M = 7 -39 CAGCGGCAT * NM:i:1
  
```

QNAME	FLAG	RNAME	POS	MAPQ	CIGAR	RNEXT	PNEXT	TLEN	SEQ	Optional fields in the
(query template name, aka, read ID)	(indicates alignment information about the read, e.g. paired, aligned, etc.)	(reference sequence name, e.g. chromosome / transcript id)	(1-based position)	(mapp quality)				(the number)	(read)	
					M	Match			Exact match of p positions	
					X	Mismatch			Next p positions are not identical	
					D	Deletion			Next p positions on ref are absent in the query	
					I	Insertion			Next p positions on query are absent in the ref, so are "inserts" in the query	
					N	Gap			Next p positions on ref are known, in advance of alignment, to be absent on the query	

CIGAR quick-ref

M	Match or mismatch (p positions)
X	Confirmed mismatch
=	Confirmed exact match
I	Insertion in query
D	Deletion from reference
N	Intron / skip region
S	Soft-clip (unmapped ends)
H	Hard-clip (trimmed)

Step 4 — Variant calling: the pileup

Once reads are aligned, the caller reads down each column and asks: **do the reads agree with the reference**, or not?

Reference	A	C	G	T	A	C	G	A	T	C
			G	T	A	C	G	A	T	C
			G	T	A	C	G	A	T	C
			G	T	A	C	G	A	T	C
			G	T	A	C	G	A	T	C
A	C	G	T	A	C	G	A			
A	C	G	T	A	C	G	A	T		
A	C	G	T	A	C	A				
A	C	G	T	A	C	A	A			
	C	G	T	A	C	A	A			

Position highlighted

6 reads: A → G

3 reads: A (ref)

Caller weighs depth (DP) and base/mapping quality before calling the SNP.

The VCF file records every variant

The caller's output is a Variant Call Format file — one line per difference from the reference.

CHROM / POS	where the variant is
REF / ALT	reference vs observed allele
QUAL	confidence in the call
DP	read depth at the site
AF	allele frequency (mixed infections)

Common callers

- **SAMtools** / BCFtools
- GATK HaplotypeCaller
- FreeBayes

SNP pipelines:

Snippy, CFSAN add bacterial-specific filters.

VCF/BCF encodes every variant relative to the reference

VCF

```
##fileformat=VCFv4.2
##contig=<ID=2,length=51304566>
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT SAMPLE1 SAMPLE2 SAMPLE3 SAMPLE4 SAMPLE5 SAMPLE6 SAMPLE7
2 81170 . C T . . AC=9;AN=7424 GT:DP:GQ 0/0:4:12 0/0:3:9 0/1:1:3 0/1:9:24 1/0:4:12 0/0:5:15 0/0:4:12
2 81171 . G A . . AC=6;AN=7446 GT:DP:GQ 0/1:4:12 0/0:3:9 0/0:1:3 0/0:9:24 0/1:4:12 0/1:5:15 0/0:4:12
2 81182 . A G . . AC=5;AN=7506 GT:DP:GQ 0/0:5:15 0/0:4:12 0/0:5:15 0/0:9:24 0/0:4:12 0/0:4:12 0/0:4:12
2 81204 . T G . . AC=2;AN=7542 GT:DP:GQ 1/0:5:15 0/0:9:27 0/0:10:30 0/0:15:39 0/0:9:27 1/0:13:39 0/1:14:42
```

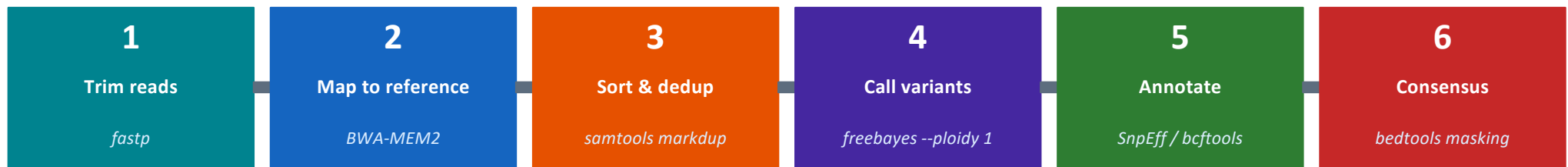
BCF

```
2 81170 . C T . . AC=9;AN=7424 GT:0/0:0/0:0/1:0/1:1/0:0/0:0/0 DP:4:3:1:9:4:5:4 GQ:12: 9: 3:24:12:15:12
2 81171 . G A . . AC=6;AN=7446 GT:0/1:0/0:0/0:0/0:0/1:0/1:0/0 DP:4:3:1:9:4:5:4 GQ:12: 9: 3:24:12:15:12
2 81182 . A G . . AC=5;AN=7506 GT:0/0:0/0:0/0:0/0:0/0:0/0:0/0 DP:5:4:5:9:4:4:4 GQ:15:12:15:24:12:12:12
2 81204 . T G . . AC=2;AN=7542 GT:1/0:0/0:0/0:0/0:0/0:0/1:0/0/1 DP:5:9:10:15:9:13:14 GQ:15:27:30:39:27:39:42
```

VCF header + records across 7 samples (GT = genotype, DP = depth, GQ = genotype quality); BCF is the binary equivalent

Snippy automates bacterial variant calling — from FASTQ to annotated VCF in one command

Snippy wraps BWA-MEM2, SAMtools, freebayes, and SnpEff to map Illumina reads against a reference genome and produce annotated SNP/indel calls with a single command. snippy-core then merges multiple samples into a core SNP alignment for phylogenetics.



Key commands

```
# Single sample
snippy --ref ref.gbk --R1 R1.fq.gz --R2 R2.fq.gz \
  --outdir sample1/ --cpus 8 --mincov 10

# Multi-sample core alignment
snippy-core --ref ref.fa sample1/ sample2/ --prefix core
```

Key output files

<code>snps.vcf</code>	All filtered SNPs & INDELs with annotation
<code>snps.bam</code>	Sorted, indexed alignment file
<code>snps.tab</code>	Human-readable SNP table (TSV)
<code>snps.consensus.fa</code>	Full-genome consensus (low depth → N)
<code>core.full.aln</code>	(snippy-core) Whole-genome alignment incl. invariants

💡 freebayes runs in haploid mode (`--ploidy 1`) — critical for bacteria. SNPs called only where ALT freq $\geq$ `--minfraction` (default 0.9) and depth $\geq$ `--mincov` (default 10×).

What read mapping does well

Fast & scalable

Hundreds of millions of reads mapped to a bacterial genome in minutes — hundreds of isolates a week.

Precise SNPs

At 30–50× coverage, per-base SNP calling is highly accurate.

Directly comparable

All isolates share one coordinate system — easy cross-sample variant matrices.

Quantitative

Allele frequencies and depth are preserved — detect mixed infections and minority variants.

Limitation 1 — Reference bias

The mapper places reads only on what the reference contains. **Any sequence absent from the reference is simply invisible to the analysis.**

- Large insertions and accessory genes
- Plasmids, prophages, integrons
- Capsule / serotype loci that differ from the reference

The pangenome problem

In *E. coli*, *S. pneumoniae* or *K. pneumoniae* the accessory genome can rival the core genome in size — so reference bias is severe.

Limitation 2 — Repeats, artefacts, contamination

Repeats —

rRNA operons and transposons map equally well to several places → **low MAPQ** → left uncalled.

PCR duplicates —

inflate apparent support; mark them (Picard, Sambamba) before calling.

Homopolymer errors —

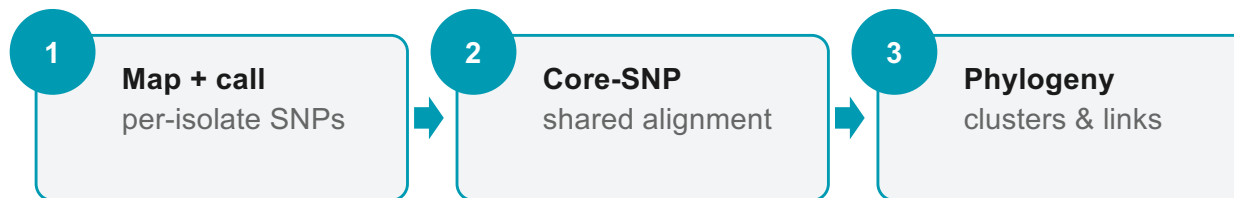
on some platforms create spurious indels.

Contamination —

reads from another organism cause false calls — pre-check with Kraken2 or alike.

Application — Outbreak investigation

Mapping each isolate and calling SNPs gives a high-confidence variant set. Shared SNPs are concatenated into a core-SNP alignment and used to build a phylogeny — SNP distances define tight clusters and suggest who infected whom.



A core-SNP tree



Typical resolution: $\leq 5-10$ SNPs between epidemiologically linked bacterial isolates of the same outbreak

Application — Antimicrobial-resistance surveillance

Resistance from point mutations

- Map to a reference allele, then call SNPs
- *gyrA* / *parC* → fluoroquinolones
- *rpoB* → rifampicin
- *katG* → isoniazid

Acquired resistance genes

- Carried on mobile elements
- SRST2 maps reads straight to gene databases
- CARD, ResFinder — no assembly needed
- Sensitive detection of known genes

Mapping answers “is this known resistance mutation/gene present?” directly from reads.

Mapping vs assembly — choose by the question

Reach for mapping when...

- A good reference exists
- You need precise SNPs fast
- Comparing many isolates to one coordinate system
- Tracking outbreaks, known AMR mutations, viruses

Reach for assembly when...

- There is no close reference
- You need the full genome content
- Typing (cgMLST), serotyping, plasmids
- Discovering accessory / new genes

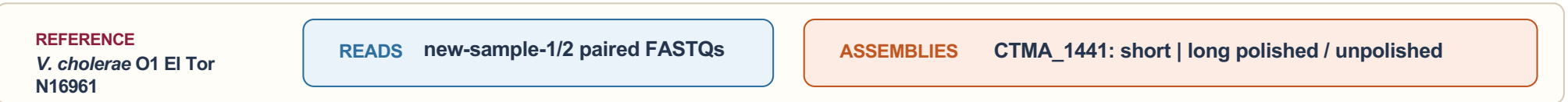
In practice, modern pipelines do both.

Key takeaways

- **Reads are anonymous** — mapping restores their genomic position
- **Indexing makes it fast** — from thousands of CPU-hours to minutes
- **The workflow is FASTQ → QC → BAM → VCF** — standard files at every step
- **Mapping excels at SNPs** — precise, comparable, quantitative; ideal for outbreaks & AMR
- **Its blind spot is the reference** — anything not in it is invisible (reference bias)

From reads to a contextualised *Vibrio cholerae* tree

Map new genomes to strain N16961, compare assembly-derived variants, then infer a rooted SNP phylogeny.



Inspect & discuss

1. Inspect snps.log, BAM, VCF and consensus outputs
2. Does polishing change the variable sites?

Interpret biology in the context of mapping quality, reference choice and assembly method.

questions?

César Rodríguez

cesar.rodriguezsanchez@ucr.ac.cr

